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PROJECT: SMART EDUCATION ASSISTANCE

THEME: SMART EDUCATION

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TABLE OF CONTENTS

Problem Statement.....	3
AI Solution.....	3
Goal of AI Solution.....	4
Process of Empathize in Design Thinking.....	4
Process of Define in Design Thinking.....	4

PROBLEM STATEMENT

Every student in traditional educational systems adheres to the same learning structure, regardless of their learning style, abilities, or areas of weakness. This "one-size-fits-all" approach creates serious problems in the classroom, where slower learners find it difficult to keep up and rapid learners may lose interest. Many students score poorly on exams and lack enthusiasm because they are unaware of their specific knowledge deficiencies until it is too late.

Additionally, teachers find it challenging to pinpoint students' areas of weakness, particularly when overseeing large classrooms. Because manual tracking of student development mostly relies on subjective observation rather than data-driven analysis, it is time-consuming and frequently erroneous.

Furthermore, most existing e-learning platforms place a higher priority on material distribution than on adaptive learning. They don't provide customised learning paths that adapt to each student's performance, and they don't track students' advancement in real time.

An AI-powered system that can automatically evaluate student performance, identify knowledge gaps, and suggest customised learning paths is therefore required. By offering real-time insights, such a system would benefit instructors and students alike, guaranteeing a more effective, interesting, and customised learning environment.

AI SOLUTION

To solve the limitations of traditional learning systems, in which the students take the same path of learning regardless of knowing about their strengths or weaknesses, we proposed the AI adaptive learning system. In this way, the system allows the students to learn more effectively by assessing their performance on tests, assignments, and learning behaviors such as time spent on the lectures and number of attempts. The system applies an AI algorithm to obtain knowledge gaps and indicate specific areas of improvement for the students.

Through this analysis, the system can automatically come up with a personalized learning path. It offers lessons, videos, tutorials, and further resources based on the student's present level of knowledge. The learning route is totally adaptive where it will update in real-time as students progress so that it keeps the focus on the most crucial topics.

The system consists of activity tracking dashboards, evaluation metrics, and a graphical knowledge map that shows the students' proficiency levels among others. With this, the instructors can have a clear view not only of the single learners but of the whole class which enables them to monitor the learners' progress and spot the common weaknesses. This approach can offer a customized, engaging, and research-based learning process that takes the student and the lecturer into account through the merging of insightful evaluation and flexible lesson planning.

GOAL OF AI SOLUTION

The main goal of this project is to improve learning efficiency and understanding by creating a system of education that is data-driven, flexible, and customized for each student. Particularly, the system aims to:

- 1. Identify and overcome knowledge gaps early.**

The system provides continuous performance tracking and intelligent data analysis. By analyzing students' quiz and test marks, the AI will analyze which topic the students are lacking and instantly calculate the target marks they need to achieve an A.

- 2. Personalize learning modules.**

After analyzing the topic the students are lacking, AI will suggest some modules or materials for the student to improve on. The AI will keep track of students' learning journey and by the end of the modules, the AI will generate a trial exam based on past year questions provided by the lecturers. This process helps students to identify their current understanding and overall performance. Therefore, it leads to more consistent and long-term understanding of their courses.

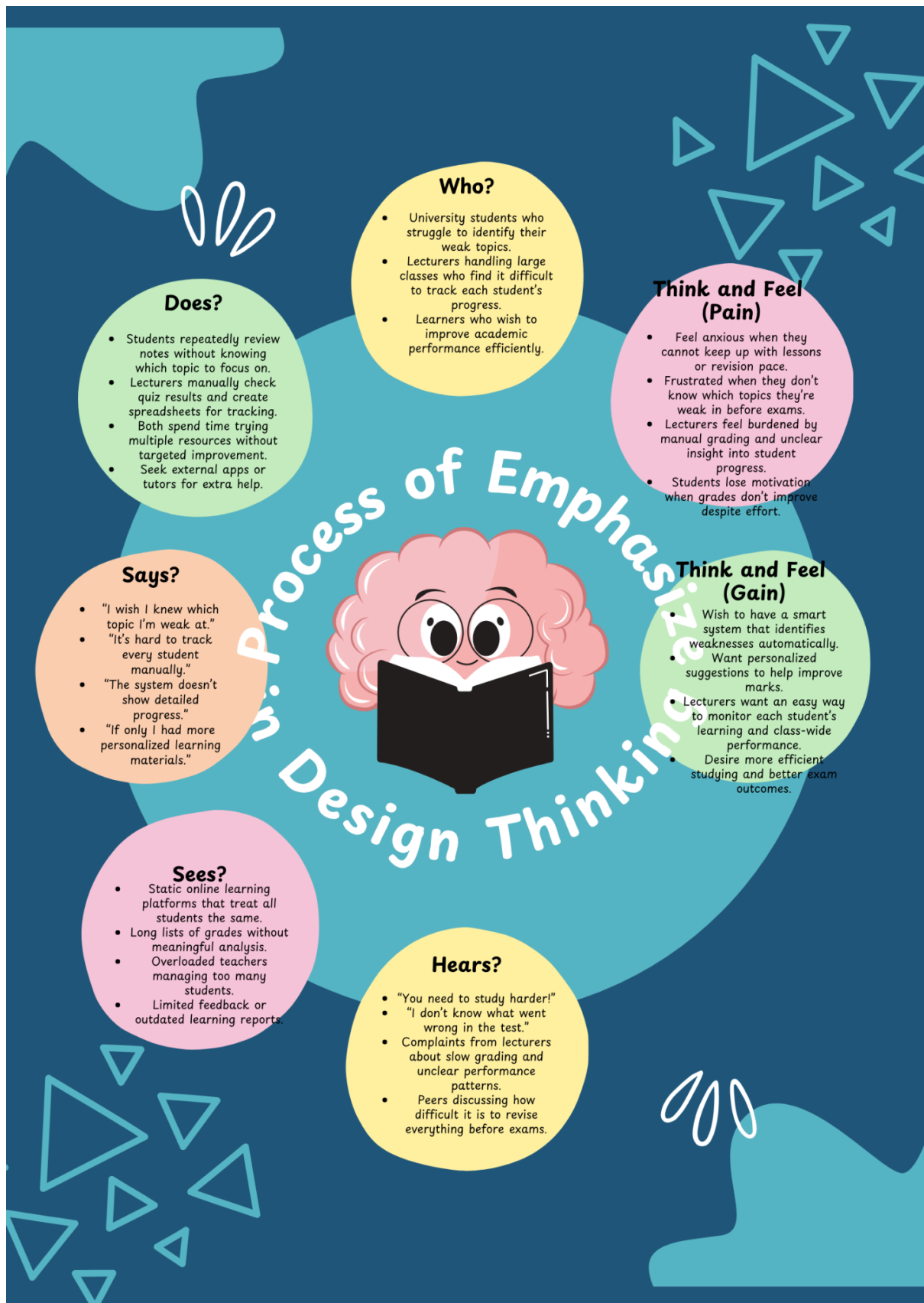
- 3. Assist lecturers in tracking each student's progress effectively.**

AI will help lecturers to track each student's progress by having an access to monitor what their students' weaknesses are, including weak topics and their effort to improve on what they are lacking. At the end of the trial exam, lecturers could identify whether their student is prepared or not for the actual final examination.

- 4. Reduce the gap between students who have difficulties in understanding and those who are advanced.**

With the help of AI, students who are struggling could work harder in their studies as this AI already identifies their weaknesses and provides some materials for them to work on so that they could get an A in their courses and be on par with the advanced students.

PROCESS OF EMPATHIZE IN DESIGN THINKING



PROCESS OF DEFINE IN DESIGN THINKING

After developing empathy for instructors and students, the team determined the fundamental problems that must be solved and precisely described the issue to direct the creation of a successful AI solution.

User Persona

- i) Student Persona: Students frequently struggle to identify their areas of weakness. To increase their comprehension, they require a system that can evaluate their performance, pinpoint their areas of weakness and provide them with tailored learning resources.
- ii) Lecturer Persona: It might be difficult for lecturers to keep track of each student's progress, particularly in large classrooms. In order to concentrate their instruction on subjects where students struggle the most, they want an AI tool that summarises pupils' strengths and weaknesses.

Problem Statement (Defined)

“Lecturers need a way to effectively monitor student progress and performance trends while students need an intelligent learning system that can automatically identify their knowledge gaps and provide a personalised learning path.”

Key Insights

- Students desire continuous feedback and adaptive learning rather than fixed study materials.
- Instructors are looking for an automated way to find the class's common weaknesses.
- An intuitive interface that shows advancement and learning gains is valued by both sides.

Design Requirements

- 1) AI algorithm for identifying knowledge gaps in test and quiz data.
- 2) Personalized learning path generator that adapts based on progress.
- 3) Lecturers can monitor students' performance analytics on this dashboard.
- 4) Interactive visualisation of progress like charts or knowledge maps.
- 5) Real-time updates that adjust educational resources based on improvements.